

Final Presentation

IEOR 115: TEAM 10

Professor Ken Goldberg

Fall 2020

Our Team



Kevin Feng

Chief Executive Officer



Yanling Liu

Chief Communications Officer



Brad Ling

Client - President of Formula
Electric at Berkeley



Rachel Lim



Alisha Laroia



Parker Nelson



Arsh Vishen



Carol Lee

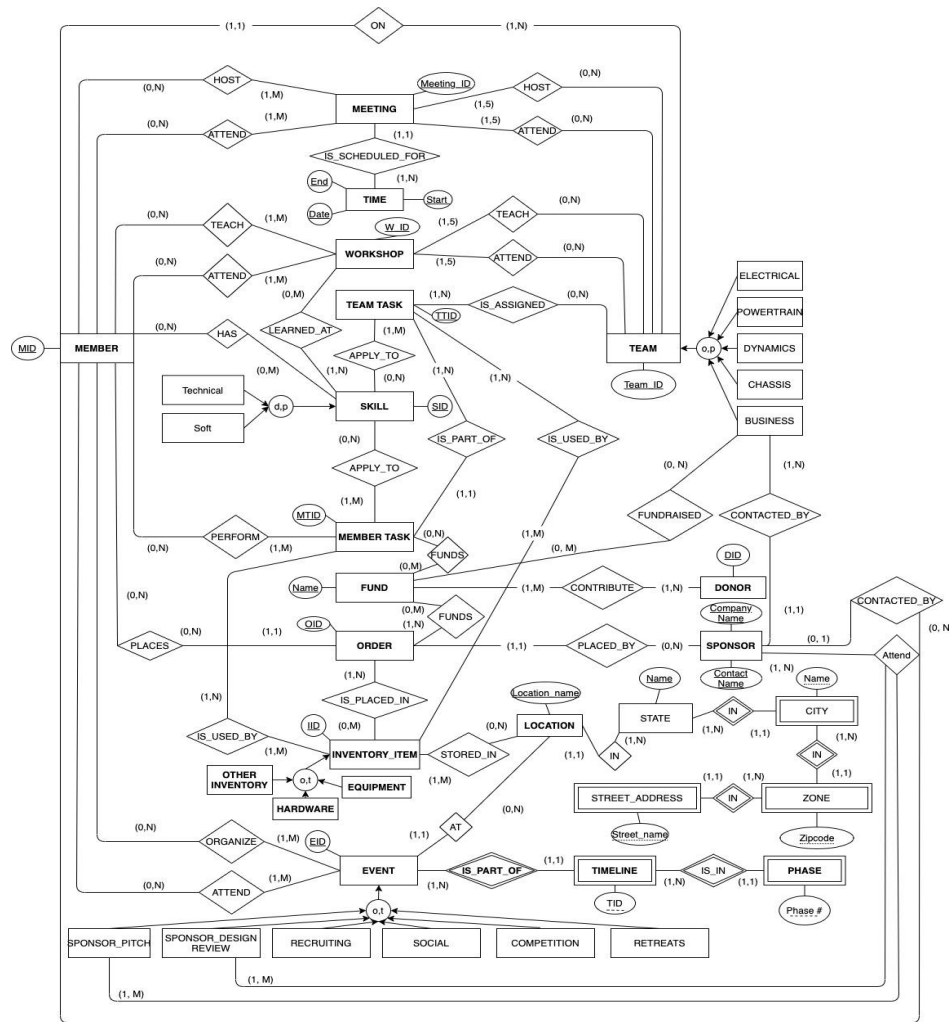


Howard Yan



Alan Huh

Simplified EER Diagram



Relational Schema

Strong Entities

1. TEAM_TASK (TTID, Due_Date, Start_Date, Description)
2. DONOR (DID, Connection, Name)
3. MEMBER_TASK (MTID, min_skill_level, Due_Date, Start_Date, Description, TTID¹)
4. INVENTORY_ITEM (IID, Name, Link, Quantity_total, Description)
 - 4a. Other_Inventory (IID⁴, manufacturer)
 - 4b. Hardware (IID⁴, brand)
 - 4c. Equipment (IID⁴, brand)
5. WORKSHOP (W_ID, Date, Time, Location, Desired_Skill)
6. LOCATION (Location_Name, State_Name²⁰)
7. ORDER (OID, Purchase_Date, Placed_By_MID¹³, Placed_By_Company_Name¹², Placed_By_Contact_Name¹²)
8. EVENT (EID, Optional, Start_date, End_date, Name, Location_Name⁴)
 - 8a. SPONSOR_PITCH (EID⁸, presenter)
 - 8b. SPONSOR_DESIGN_REVIEW (EID⁸)
 - 8c. RECRUITING (EID⁸)
 - 8d. SOCIAL (EID⁸, budget)
 - 8e. COMPETITION (EID⁸, speed_test_result, reliability_test_result)
 - 8f. RETREAT (EID⁸, budget)

Relational Schema, continued

Strong Entities

9. SKILL (SID, Name)

9a. TECHNICAL (SID⁹, Proficiency_level)

9b. SOFT (SID⁹)

10. MEETING (Meeting_ID, Meeting_Name, Type, Start¹¹, End¹¹, Date¹¹)

11. TIME (Start, End, Date)

12. SPONSOR (Company_Name, Location, Description, Connection_Note, Contact_Person_Name, Contact_Email, Contact_Number, Contacted_by_Team_ID^{14e}, Contacted_by_MID¹³)

13. MEMBER (MID, Join_date, Current/Past_Member, Role, School_Year, CSWA#, Club_Years, Team_ID¹⁴)

14. TEAM (Team_ID)

14a. ELECTRICAL (Team_ID¹⁴, car_component)

14b. POWERTRAIN (Team_ID¹⁴, car_component, design_model)

14c. DYNAMICS (Team_ID¹⁴, car_component, design_model)

14d. CHASSIS (Team_ID¹⁴, car_component, design_model)

14e. BUSINESS (Team_ID¹⁴)

15. FUND (Name)

16. STATE (Name)

Relational Schema, continued

Multivalued Attributes

- 17. ITEM (OID⁷, Item)
- 18. RECRUITING_TEAM (EID^{8c}, Recruiting_team)
- 19. DRIVER_NAME (EID^{8e}, Driver_name)

Weak Entities

- 20. CITY (Name, State_name¹⁶)
- 21. ZONE (Zipcode, City_Name²⁰, State_name¹⁶)
- 22. STREET_ADDRESS (Street_Name, Zipcode²¹, City_Name²⁰, State_name¹⁶)
- 23. TIMELINE (TID, EID⁸, Name)
- 24. PHASE (Phase#, TID²³, EID⁸, Start_date, End_date, Completed_or_not)

Relational Schema, continued

N:M Relationships

- 25. TEAM_MEETING_ATTEND (Meeting_ID¹⁰, Team_ID¹⁴)
- 26. MEMBER_MEETING_ATTEND (Meeting_ID¹⁰, MID¹³, Present)
- 27. TEAM_MEETING_HOST (Meeting_ID¹⁰, Team_ID¹⁴)
- 28. MEMBER_MEETING_HOST (Meeting_ID¹⁰, MID¹³)
- 29. WORKSHOP_MEMBER_TEACH (MID¹⁰, W_ID⁵)
- 30. WORKSHOP_MEMBER_ATTEND (MID¹⁰, W_ID⁵)
- 31. WORKSHOP_TEAM_TEACH (Team_ID¹⁴, W_ID⁵)
- 32. WORKSHOP_TEAM_ATTEND (Team_ID¹⁴, W_ID⁵)
- 33. HAS_SKILL (MID¹⁰, SID⁹)
- 34. SKILL_APPLY_TO_TASK (MTID³, SID⁹)
- 35. SKILL_LEARNED_AT (SID⁹, W_ID⁵)
- 36. MEMBER_has_MEMBER_TASK (MID¹⁰, MTID³)
- 37. TASK_ASSIGN (MID¹⁰, MTID³)
- 38. TEAM_has_TASK_ASSIGNED (TTID¹, Team_ID¹⁴)
- 39. EVENT_ATTEND (EID⁸, MID¹⁰)
- 40. MEMBER_organize_EVENT (EID⁸, MID¹⁰)
- 41. SPONSOR_attend_SPONSOR_PITCH (Sponsor_Company_Name¹², EID^{8a})
- 42. SPONSOR_attend_SPONSOR_DESIGN_REVIEW (Sponsor_Company_Name¹², EID^{8b})
- 43. MEMBER_TASK_has_FUND (MTID³, Name¹⁵)
- 44. ORDER_has_FUND (OID⁷, Name¹⁵)
- 45. TEAM_fundraised_FUND (Team_ID^{14e}, Fund_Name¹⁵)
- 46. DONOR_contributes_FUND (DID², Fund_Name¹⁵)
- 47. INVENTORY_ITEM_used_by_MEMBER (IID⁴, MTID³, date, quantity)
- 48. INVENTORY_ITEM_placed_in_ORDER (OID⁷, IID⁴, date, quantity)
- 49. INVENTORY_ITEM_has_TEAM (IID⁴, TTID¹, date)
- 50. INVENTORY_ITEM_stored_at_LOCATION (IID⁴, Location_Name⁶)

Normalization Analysis

4. INVENTORY_ITEM (IID, Name, Link, Quantity_total, Description)

- Dependencies $\{\{\text{IID}\} \rightarrow \{\text{Name, Link, Quantity_total, description}\}, \{\text{Link}\} \rightarrow \{\text{Name, Description}\}\}$
 - Link is unique, Name is generic (i.e. screws, glue, etc)
- 2NF $\rightarrow$ 3NF:
 - R1(IID, Link, Quantity_total)
 - R2(Link, Name, Description)

12. SPONSOR (Company_Name, Location, Description, Connection_Note, Contact_Person_Name, Contact_Email, Contact_Number, Contacted_by_Team_ID^{14e}, Contacted_by_MID¹³)

- Dependencies $\{\{\text{Company_name}\} \rightarrow \{*\}, \{\text{Contact_email}\} \rightarrow \{\text{Contact_Person_Name}\}, \{\text{Contacted_by_MID}^{13}\} \rightarrow \{\text{Contacted_by_Team_ID}^{14e}\}\}$
- 2NF $\rightarrow$ 3NF:
 - R1(Company_name, Location, Description, Connection_Note, Contact_Email, Contact_Number, Contacted_by_MID¹³)
 - R2(Contact_Email, Contact_Person_Name)
 - R3(Contacted_by_MID¹³, Contacted_by_Team_ID^{14e})

24. PHASE (Phase#, TID²³, EID⁸, Start_date, End_date, Completed_or_not)

- Dependencies $\{\{\text{Phase\#, TID}^{23}, \text{EID}\} \rightarrow \{\text{Start_date}\}, \{\text{Phase\#, TID}^{23}, \text{EID}\} \rightarrow \{\text{End_date}\}, \{\text{Phase\#, TID}^{23}, \text{EID}\} \rightarrow \{\text{Completed_or_not}\}, \{\text{End_date}\} \rightarrow \{\text{Completed_or_not}\}\}$
- 2NF $\rightarrow$ 3NF:
 - R1(Phase#, TID²³, EID⁸, Start_date, End_date)
 - R2(End_date, Completed_or_not)

Query 1: Evaluate metrics to aid member recruitment

- What is the count of members in each role?
- What is the distribution of grade levels and majors within the organization?
- Benefits:
 - This will help FEB analyze whether there are not enough members in a certain role, which can help FEB make recruiting decisions in future semesters (e.g. taking more members who are interested in a certain role or moving members to fill roles). In other words, this will help FEB gauge and distribute manpower.
- Method: Generate time series plots for analysis

Queries 2 & 3: Evaluate Existing / Prospective Donors and Sponsors

- Determine which donors contributed the most funding
- Determine which sponsors are the most actively engaged
- Identify which spare parts sponsors contributed and tailor requests according to inventory needs
- Determine future outreach efforts based on this information
- Benefit:
 - Understanding the extent of contribution from existing donors and sponsors.
 - Knowing which donors and sponsors to actively be in contact with.
 - Identify most enthusiastic industries for sponsors and donors
 - Estimate future donations from donors

Query 4: Evaluate each member's performance every semester

- 4 Criteria: Participance, Leadership, Public Relations, Skill set/level
 - **Participation:** How often do members participate in workshops? How active they are in teaching and guiding other members?
 - **Leadership:** What workshops has the member led?
 - **Public Relations:** Is the member active in seeking sponsorships? Have the members obtained any orders from sponsors?
 - **Skill Set/level:** How much a member has grown through skill trainings?
- Business Justification:
 - Identify potential leaders among members that could be encouraged to apply for leadership positions
- Benefits:
 - Highlight members by predicted score based on computed features
 - Assist evaluating candidates that participate in elections

Query 4: Synthetic Data

- Client gave us sample data to use – team roster, sponsors, events
- Altered any sensitive information by hand and uploaded data to MySQL by .csv
- Created data for these tables:
 - EVENT
 - EVENT_ATTENDED
 - HAS_SKILL
 - INVENTORY_ITEM
 - MEETING
 - MEMBER
 - MEMBER_HAS_MEMBER_TASK
 - MEMBER_MEETING_ATTEND
 - MEMBER_ORGANIZE_EVENT
 - MEMBER_TASK
 - ORDERS
 - SKILL
 - SOFT
 - TECHNICAL
 - SKILL_LEARNED_AT
 - SPONSOR
 - TEAM
 - TEAM_TASK
 - TIME
 - WORKSHOP
 - WORKSHOP_MEMBER_ATTEND
 - WORKSHOP_MEMBER_TEACH

EID	Optional	Start_date	End_date	Name	Location_name
1001	0	2018-06-07	2018-06-14	COMPETITION	San Jose
1002	0	2019-06-07	2019-06-14	COMPETITION	San Jose
1003	0	2020-06-07	2020-06-14	COMPETITION	San Jose
1004	1	2019-01-23	2019-02-02	RECRUITING	Berkeley
1005	1	2019-08-17	2019-08-23	RECRUITING	Berkeley
1006	1	2020-01-24	2020-02-03	RECRUITING	Berkeley
1007	0	2019-01-23	2019-02-02	RETREAT	Monterey
1008	0	2019-08-17	2019-08-23	RETREAT	Oakland
1009	0	2020-01-24	2020-02-03	RETREAT	Napa
1010	1	2019-02-08	2019-02-09	SOCIAL	Berkeley
1011	1	2019-09-13	2019-09-14	SOCIAL	Berkeley
1012	1	2020-02-07	2020-02-08	SOCIAL	Berkeley
1013	0	2019-02-21	2019-02-21	SPONSOR_...	Berkeley
1014	0	2019-09-24	2019-09-24	SPONSOR_...	Berkeley
1015	0	2020-02-14	2020-02-14	SPONSOR_...	Berkeley
1016	0	2019-02-11	2019-02-11	SPONSOR_P...	Berkeley
1017	0	2019-09-09	2019-09-09	SPONSOR_P...	Berkeley
1018	0	2020-02-05	2020-02-05	SPONSOR_P...	Berkeley

Query 4: SQL Queries Measuring Participation

Attribute	Explanation	Code
#_WORKSHOP_ATTEND	# of workshops each member attended.	SELECT MEMBER_MID, COUNT(WORKSHOP_W_ID) AS "NUM_WORKSHOP_ATTENDED" FROM WORKSHOP_MEMBER_ATTEND GROUP BY MEMBER_MID;
#_TASK_PERFORM	# of tasks each member has performed.	SELECT MEMBER_MID, COUNT(MEMBER_TASK_MTID) AS TASK_PERFORMED FROM MEMBER_has_MEMBER_TASK GROUP BY MEMBER_MID;
MEETING_ATTENDANCE	% of meetings a member was present at.	SELECT m.MID, (a.Mts_Attended / a.Total_Mts) AS MEETING_ATTENDANCE FROM 'MEMBER' m LEFT JOIN (SELECT MEMBER_MID, COUNT(MEETING_Meeting_ID) AS Total_Mts, SUM(Present) as Mts_Attended FROM MEMBER_MEETING_ATTEND GROUP BY MEMBER_MID) AS a ON m.MID = a.MEMBER_MID;
EVENT_ATTENDANCE	% of events each member attended since they joined the club.	SELECT a.MID, (a.EVENT_ATTENDED / b.NUM_EVENT) AS EVENT_ATTENDANCE FROM (SELECT m.MID, COUNT(ea.EVENT_EID) as EVENT_ATTENDED FROM 'MEMBER' m LEFT JOIN MEMBER_attend_EVENT ea ON m.MID=ea.MEMBER_MID GROUP BY m.MID) AS a JOIN (SELECT m.MID, COUNT(e.EID) as NUM_EVENT FROM 'MEMBER' m LEFT JOIN EVENT e ON e.Start_date >= m.Join_date GROUP BY m.MID) AS b ON a.MID=b.MID;

First 10 Rows of Output:

#_WORKSHOP_ATTEND

MEMBER_MID	NUM_WORKSHOP_ATTENDED
1	3
2	1
3	1
4	1
5	1
7	1
8	2
9	2
10	1

#_TASK_PERFORM

MEMBER_MID	TASK_PERFORMED
1	1
2	1
3	3
7	2
8	1
9	3
11	1
13	1
15	4
16	1

MEETING_ATTENDANCE

MID	MEETING_ATTENDANCE
1	1
2	1
7	NULL
10	1
17	NULL
27	0.75
33	1
35	NULL
3	1
11	1

EVENT_ATTENDANCE

MID	EVENT_ATTENDANCE
1	0.2222
2	0.2353
3	0.5455
4	0.1765
5	0.1176
6	0.5000
7	0.1667
8	0.1667
9	0.1176
10	0.0588

Query 4: SQL Queries Measuring Leadership, PR, & Skills

Attribute	Explanation	Code
#_WORKSHOP_AND_EVENT_HOSTED	# of workshops a member has taught and the number of events a member has organized.	<pre>SELECT a.MEMBER_MID, COUNT(a.WORKSHOP_W_ID) as WORKSHOP_TAUGHT, COUNT(a.EVENT_EID) as EVENT_ORGANIZED FROM(SELECT wt.MEMBER_MID, wt.WORKSHOP_W_ID, eo.EVENT_EID FROM WORKSHOP_MEMBER_TEACH as wt LEFT JOIN MEMBER_organize_EVENT as eo ON wt.MEMBER_MID=eo.MEMBER_MID UNION ALL SELECT eo.MEMBER_MID, wt.WORKSHOP_W_ID, eo.EVENT_EID From WORKSHOP_MEMBER_TEACH as wt RIGHT JOIN MEMBER_organize_EVENT as eo On wt.MEMBER_MID=eo.MEMBER_MID WHERE wt.MEMBER_MID is null) as a GROUP BY a.MEMBER_MID;</pre>
#_SPONSOR_ORDERS_OBTAINED	# of sponsored orders per company a member helped get.	<pre>SELECT m.MID, s.Company_Name, COUNT(o.oid) AS No_Sponsor_Orders_Obtained FROM SPONSOR as s INNER JOIN 'MEMBER' as m ON s.Contacted_by_MID = m.MID INNER JOIN 'ORDER' as o ON s.Company_Name = o.Placed_By_Company_Name GROUP BY m.MID, s.Company_Name ORDER BY COUNT(o.oid);</pre>
#_SKILLS_LEARNED	# of skills each member has learned from workshops taught by the club.	<pre>SELECT hs.MEMBER_MID, COUNT(hs.SKILL_SID) AS SKILLS_LEARNED FROM HAS_SKILL as hs INNER JOIN SKILL_LEARNED_AT as learn ON hs.SKILL_SID = learn.SKILL_SID GROUP BY hs.MEMBER_MID ORDER BY COUNT(hs.SKILL_SID) DESC;</pre>

First 10 Rows of Output:

#_WORKSHOP_AND_EVENT_HOSTED

MEMBER_MID	WORKSHOP_TAUGHT	EVENT_ORGANIZED
1	2	2
3	3	3
4	2	2
7	3	3
8	1	1
9	2	0
10	1	1
12	1	0
13	2	0
14	1	0

#_SKILLS_LEARNED

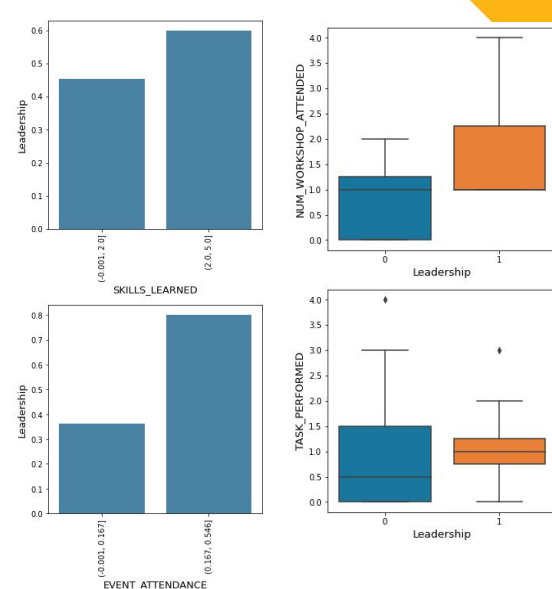
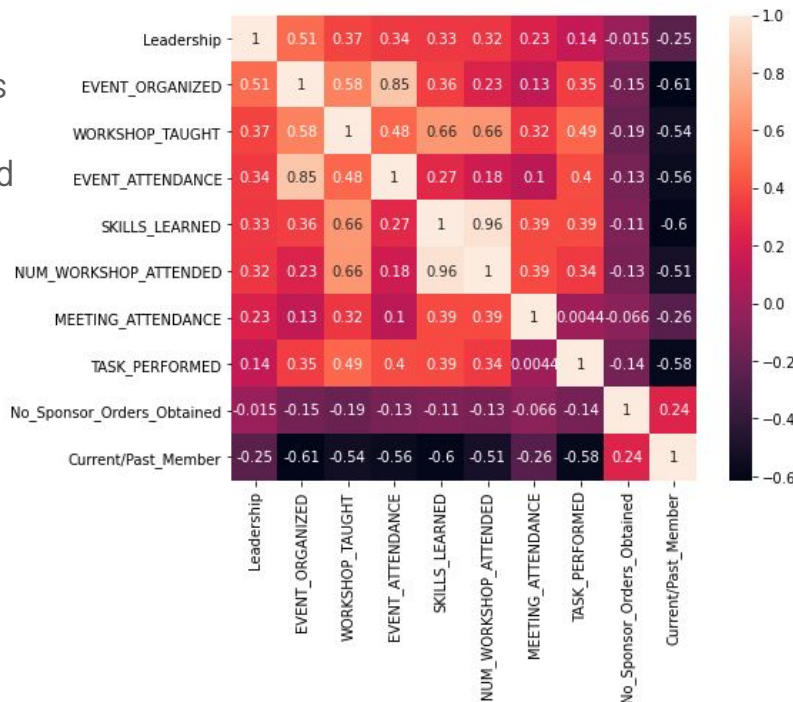
MEMBER_MID	SKILLS_LEARNED
1	5
13	5
8	4
12	3
9	3
2	2
10	2
14	2
23	2
3	2

#_SPONSOR_ORDERS_OBTAINED

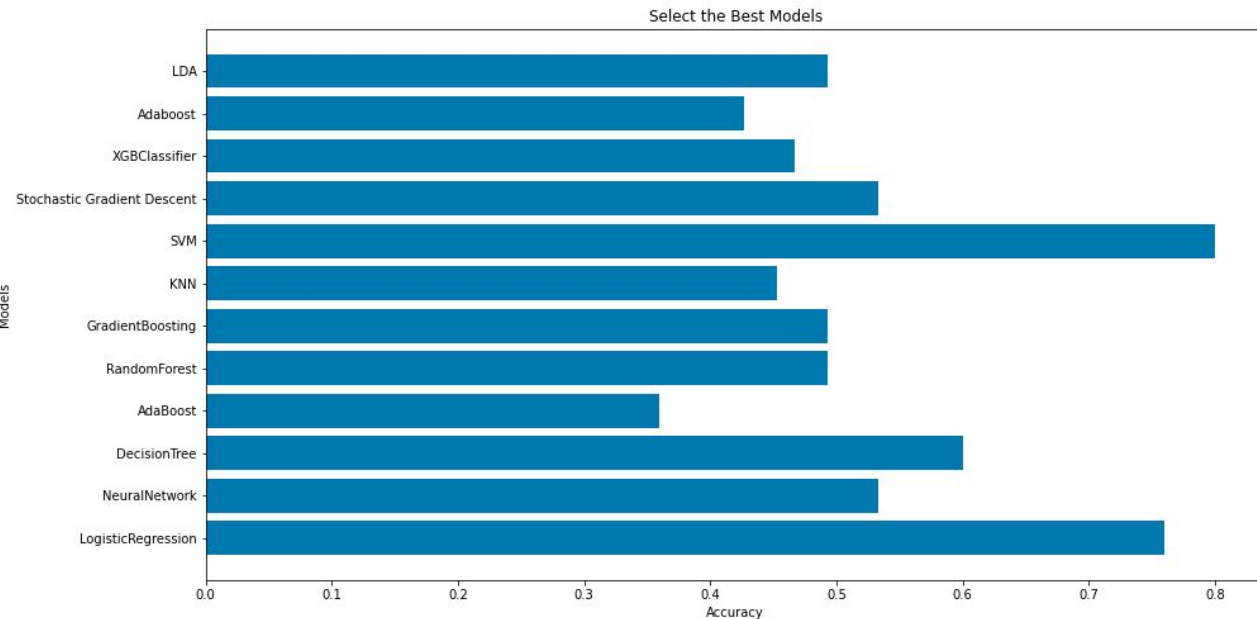
MID	Company_Name	No_Sponsor_Orders_Obtained
18	CNC Solutions	1
18	East Bay Machine	1
18	Hawk CNC	1
18	Marx Digital Manufacturing	1
19	Alco Metals	1
19	Bob McGee's Machining Company	1
19	Clancy Machine Tool, Inc	1
19	Electric Motorsport Inc	1
20	CalPol CNC Machining	1
20	High Speed CNC	1

Query 4: EDA

- Desirable correlations
- No_Sponsor_Obtained may be omitted
- Difference in features for students with leadership and non-leadership positions



Query 4: Model Selection



```
#Select the best models
random_state = 107
clfs = []
clfs.append(LogisticRegression(random_state = random_state))
clfs.append(MLPClassifier(random_state=random_state))
clfs.append(DecisionTreeClassifier(random_state=random_state))
clfs.append(AdaBoostClassifier(random_state=random_state, learning_rate=0.1))
clfs.append(RandomForestClassifier(random_state=random_state))
clfs.append(GradientBoostingClassifier(random_state=random_state))
clfs.append(KNeighborsClassifier())
clfs.append(SVC(random_state=random_state))
clfs.append(SGDClassifier('log', random_state=random_state, shuffle=True))
clfs.append(XGBClassifier(random_state=random_state))
clfs.append(AdaBoostClassifier(
    DecisionTreeClassifier(random_state=random_state),
    random_state=random_state, learning_rate=0.1))
clfs.append(LinearDiscriminantAnalysis())
results = []
for clf in clfs :
    results.append(cross_val_score(clf, X_train, Y_train,
                                   scoring = "f1", cv = 5, n_jobs=-1))

cv = [x.mean() for x in results]
clf_name = ["LogisticRegression", "NeuralNetwork", "DecisionTree",
            "AdaBoost", "RandomForest", "GradientBoosting", "KNN", "SVM",
            "Stochastic Gradient Descent", "XGBClassifier", "Adaboost", "LDA"]
plt.figure(figsize=(15, 8))
plt.ylabel('Models')
plt.xlabel('Accuracy')
plt.title('Select the Best Models')

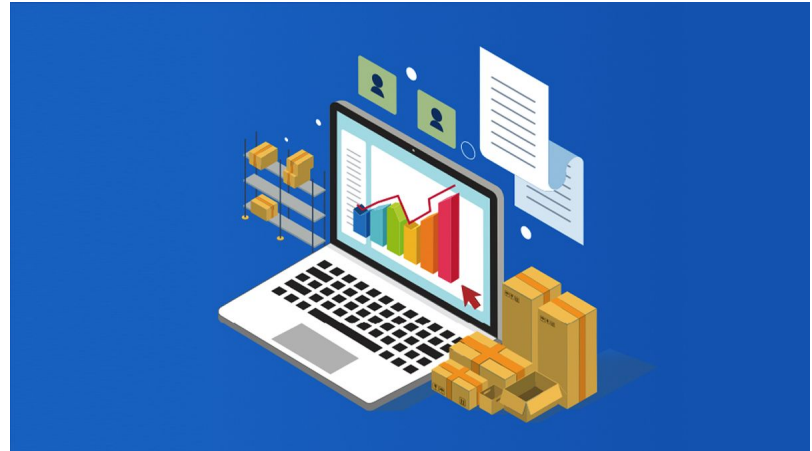
plt.barh(clf_name, cv);
```


Query 4: Model & Output

	Support Vector Machine	Logistic Regression																																																						
Idea	- Leadership Scores = Weighted sum of inputs - Rank members for leadership potential based on predicted scores																																																							
Function	$\mathbf{w}^T \mathbf{x} + b = 0$ $\mathbf{w}^T \mathbf{x} + b \geq 0 \text{ for } d_i = +1$ $\mathbf{w}^T \mathbf{x} + b < 0 \text{ for } d_i = -1$	$h(x) = \frac{1}{1 + e^{-\theta^T x}} \quad h(x) = \begin{cases} > 0.5, & \text{if } \theta^T x > 0 \\ < 0.5, & \text{if } \theta^T x < 0 \end{cases}$																																																						
Model	<pre>svm = SVC(C = 300, gamma = 0.001, kernel = 'linear')</pre>	<pre>LR = LogisticRegression(C: 1438.44988828766, penalty: 'l1', solver: 'liblinear')</pre>																																																						
Output	<table> <thead> <tr> <th></th><th>MID</th><th>svm_score</th></tr> </thead> <tbody> <tr> <td></td><td>3</td><td>25 -3.082918</td></tr> <tr> <td>5</td><td>27</td><td>-2.425413</td></tr> <tr> <td>1</td><td>23</td><td>-2.991010</td></tr> <tr> <td>4</td><td>26</td><td>-2.995262</td></tr> <tr> <td>0</td><td>20</td><td>-3.082918</td></tr> <tr> <td></td><td>6</td><td>30 -3.710632</td></tr> <tr> <td></td><td>10</td><td>18 -4.953305</td></tr> <tr> <td></td><td>15</td><td>34 -4.953305</td></tr> </tbody> </table>		MID	svm_score		3	25 -3.082918	5	27	-2.425413	1	23	-2.991010	4	26	-2.995262	0	20	-3.082918		6	30 -3.710632		10	18 -4.953305		15	34 -4.953305	<table> <thead> <tr> <th></th><th>MID</th><th>lr_score</th></tr> </thead> <tbody> <tr> <td></td><td>3</td><td>25 0.082637</td></tr> <tr> <td>5</td><td>27</td><td>0.149503</td></tr> <tr> <td>4</td><td>26</td><td>0.112178</td></tr> <tr> <td>1</td><td>23</td><td>0.099674</td></tr> <tr> <td>0</td><td>20</td><td>0.082637</td></tr> <tr> <td></td><td>6</td><td>30 0.049273</td></tr> <tr> <td></td><td>10</td><td>18 0.014380</td></tr> <tr> <td></td><td>15</td><td>34 0.014380</td></tr> </tbody> </table>		MID	lr_score		3	25 0.082637	5	27	0.149503	4	26	0.112178	1	23	0.099674	0	20	0.082637		6	30 0.049273		10	18 0.014380		15	34 0.014380
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Next Step	Obtain more members' data, Ensembling																																																							

Query 5: Order Forecasting System

- Predict the usage of inventory items given historical member usage data
- Business Justification
 - Minimize supply and material holding costs → enable **optimal inventory management, minimal costs, and minimal probability of stockout.**



Query 5: Synthetic Data

Created data for these tables:

- TEAM
- MEMBER
- INVENTORY_ITEM
- **INVENTORY_ITEM_used_by_MEMBER**

With these items:

- Gas
- Screws
- Glue
- Latex Gloves
- Masks

	INVENTORY_ITEM_IID	MEMBER_MID	Date	Quantity
0	12	238	2017-05-14	26
1	12	261	2016-01-31	40
2	12	210	2015-02-15	37
3	12	259	2016-01-17	42
4	12	253	2016-05-01	11
...	...	...	...	...
406	8	296	2016-05-08	6
407	8	272	2017-03-19	2
408	8	279	2016-04-10	6
409	8	298	2015-01-25	4
410	8	288	2019-01-06	2

Query 5: SQL Query and Output

```
SELECT INVENTORY_ITEM_used_by_MEMBER.Date, SUM(INVENTORY_ITEM_used_by_MEMBER.Quantity)
FROM INVENTORY_ITEM_used_by_MEMBER, INVENTORY_ITEM
WHERE INVENTORY_ITEM.Name = 'Screws'
AND INVENTORY_ITEM.IID = INVENTORY_ITEM_used_by_MEMBER.INVENTORY_ITEM_IID
GROUP BY INVENTORY_ITEM_used_by_MEMBER.Date;
```

Dirty Dataframe

	Date	SUM(INVENTORY_ITEM_used_by_MEMBER.Quantity)
0	2020-05-31	21.0
1	2017-01-15	62.0
2	2020-04-26	10.0
3	2019-03-03	41.0
4	2019-04-07	9.0
...	...	...
63	2015-03-15	1.0
64	2020-01-19	1.0
65	2015-03-01	15.0
66	2015-05-03	23.0
67	2016-01-10	4.0

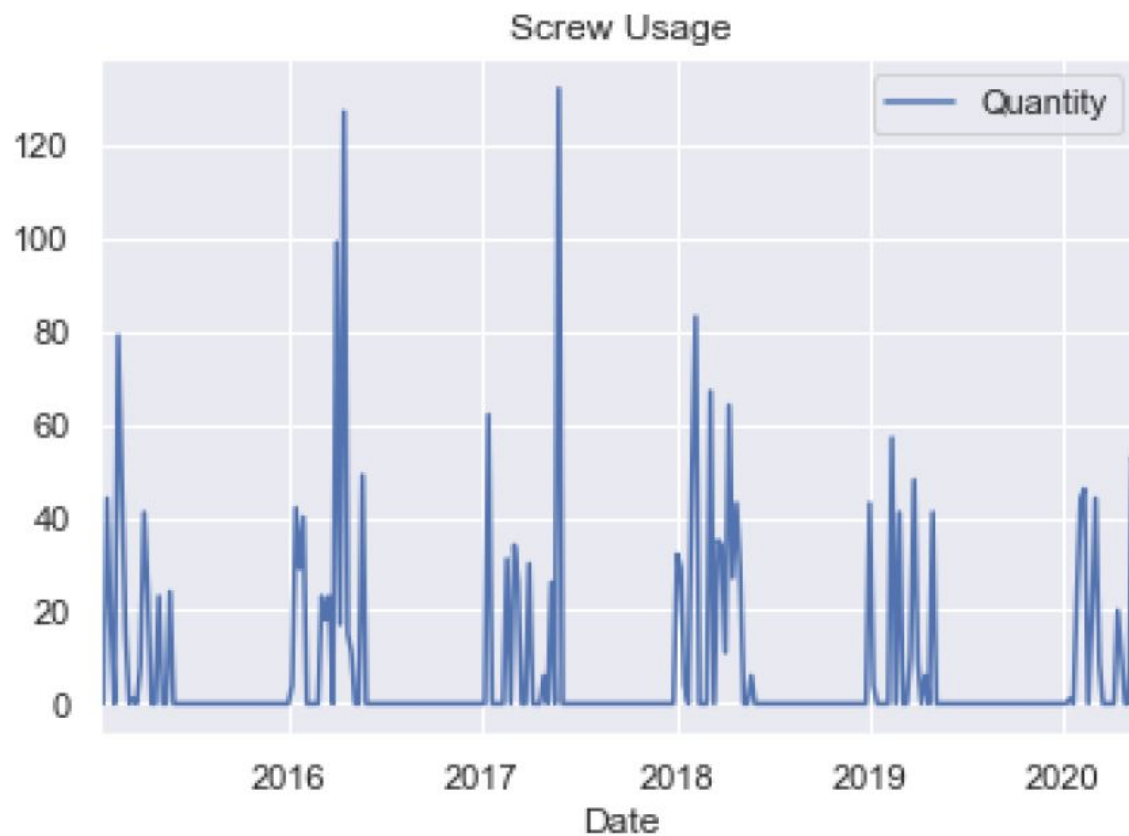
68 rows × 2 columns

Clean Dataframe

	Quantity
Date	
2015-01-11	23.0
2015-01-18	0.0
2015-01-25	44.0
2015-02-01	17.0
2015-02-08	0.0
...	...
2020-05-03	0.0
2020-05-10	0.0
2020-05-17	53.0
2020-05-24	48.0
2020-05-31	21.0

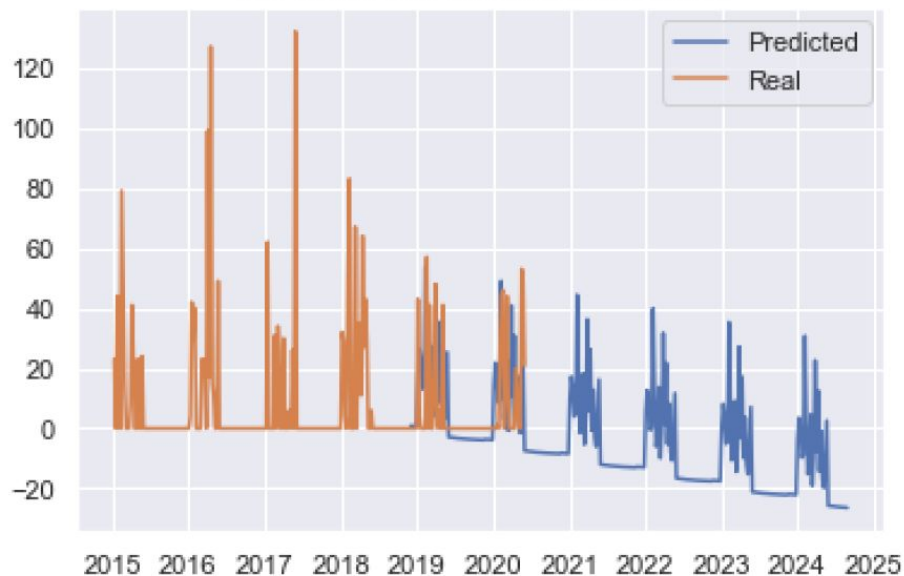
282 rows × 1 columns

Query 5: Assumptions and Models

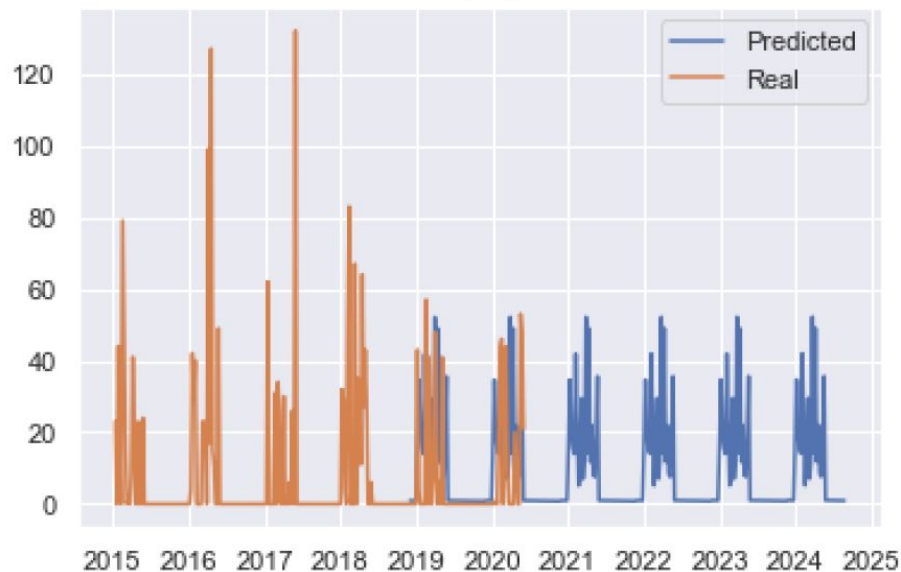


Query 5: Assumptions and Models

Self Optimized Exponential Smoothing

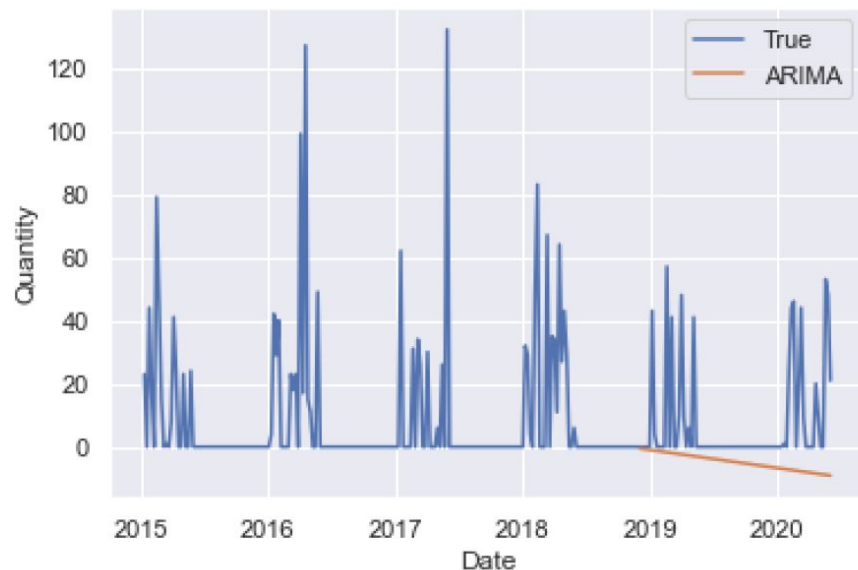


Selected Exponential Smoothing

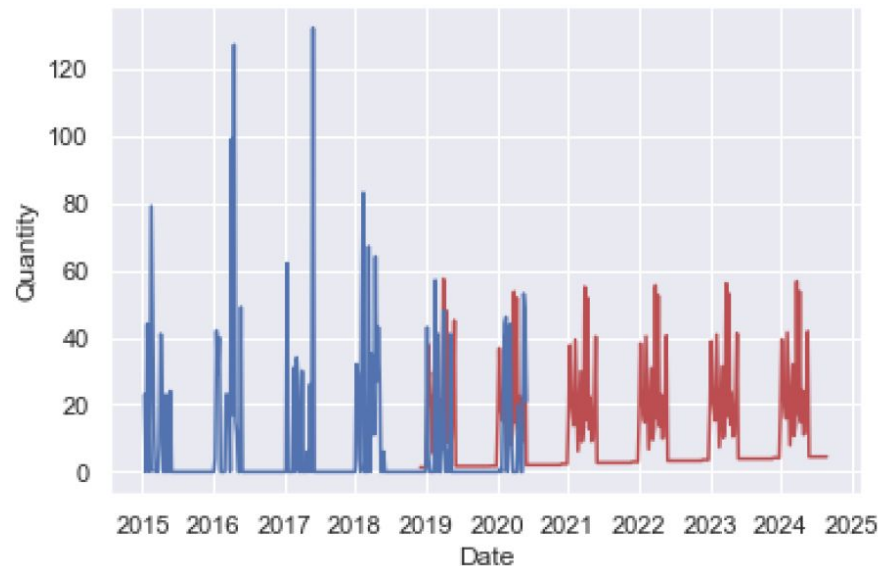


Query 5: Assumptions and Models

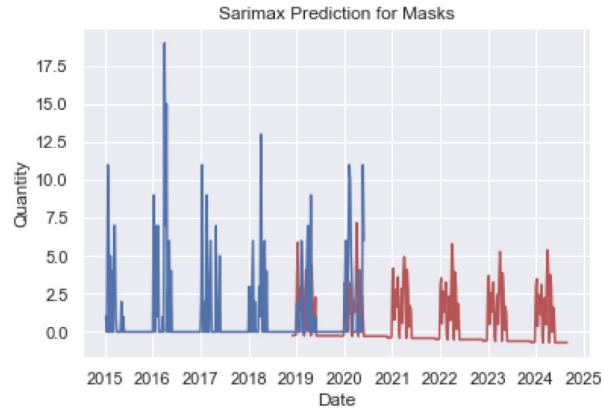
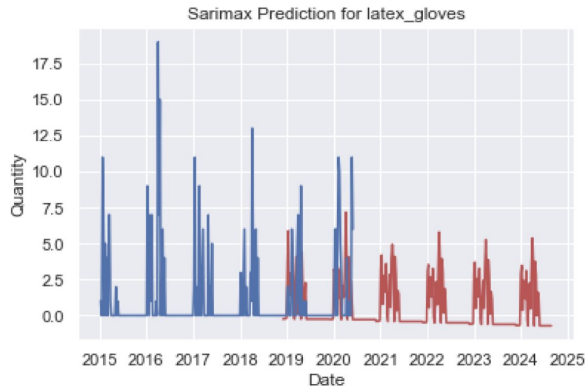
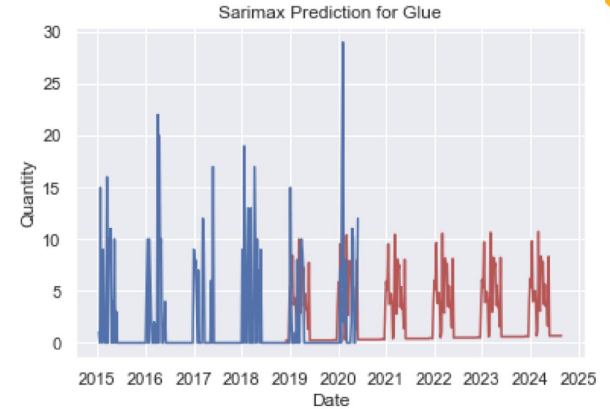
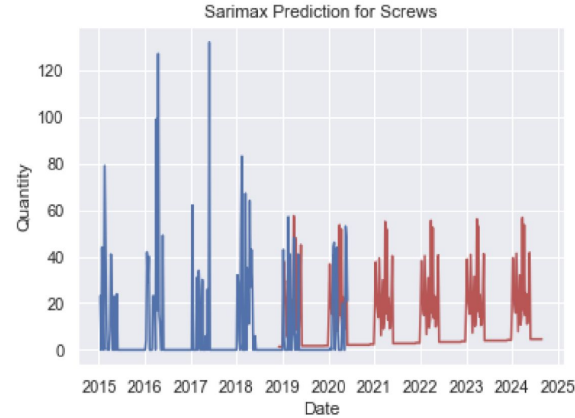
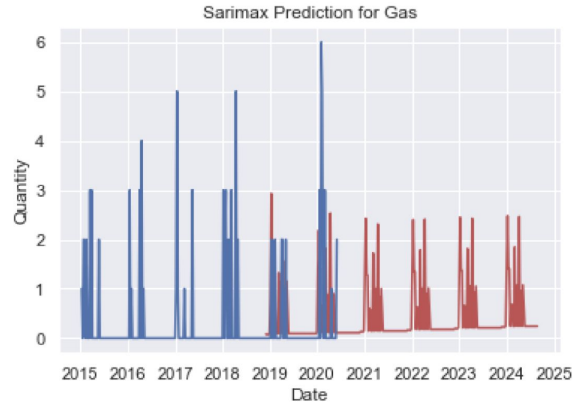
ARIMA Prediction vs True Data



SARIMA Prediction vs True Data



Query 5: Exporting Data and Plots





Thank you! Any questions?